Plan

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Four lectures on the MatterMost paper:

- The theory behind the MatterMost paper (three lectures)
- The main idea of the MatterMost paper (one lecture)

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Today:

- The theory behind the MatterMost paper 1

The theory behind the paper 1

Finite horizon sequential decision problems (SDPs)

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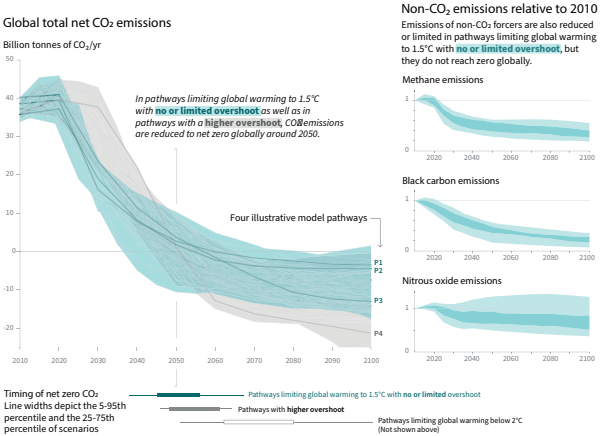
- SDPs (intertemporal optimization problems) are problems in which decisions have to be taken in a sequence, one after the other
- We look at two examples

Example 1: emission reduction (climate mitigation) policies

- Global GHG emissions must reach net-negative by about 2050



IPCC Special Report - Global Warming of 1.5 °C, Oct. 2018

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- Technologies that allow emission reductions at low costs may become available soon
- Rules and regulations may not be implemented or they may be implemented with delays

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- How to make good plans for the next few decades?
- Which plans are good under uncertainty?
- How safe is the corridor recommended by the IPCC Special Report?

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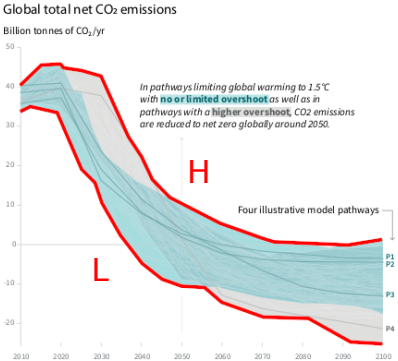
- What are the odds of paths along the boundaries of the emissions corridor?



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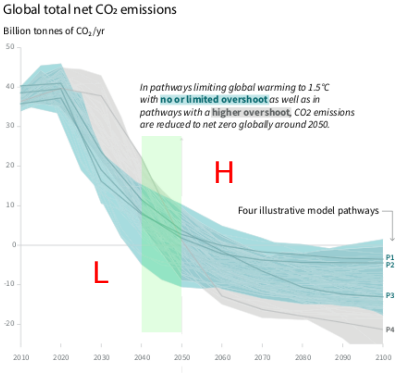
- If new technologies to reduce GHG emissions become available around 2050 . . .



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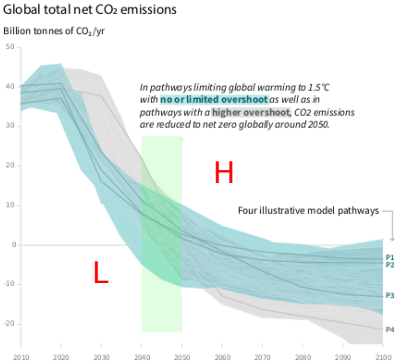

- . . . what would optimal emission plans look like?

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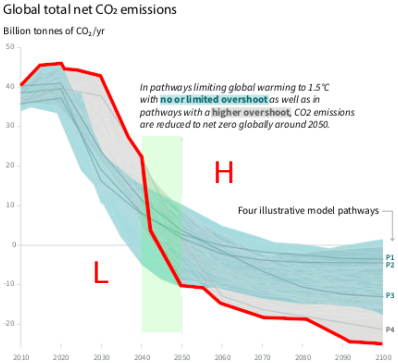

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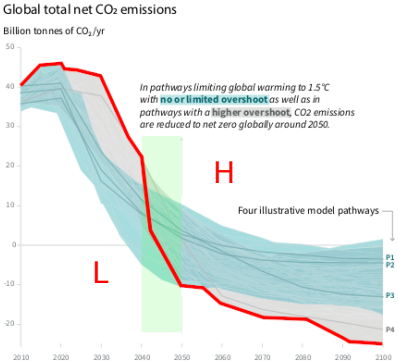

- But is this an optimal emission plan? In which sense?

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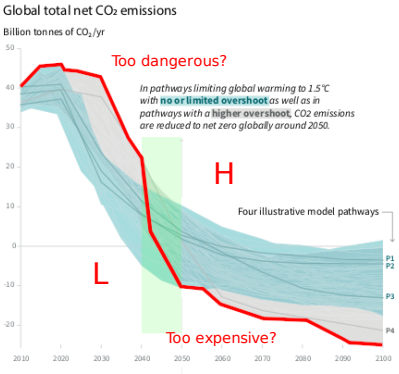

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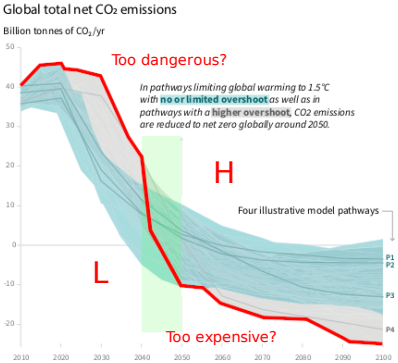

- Are L emissions after 2050 possibly too expensive?

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- BT is a bad but temporary state
- In BT, societies are poor but it is certain that the next state will be good and safe: GS

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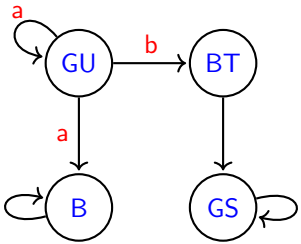

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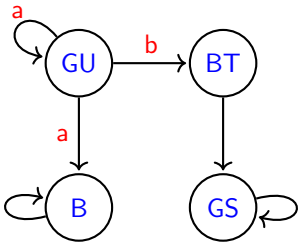
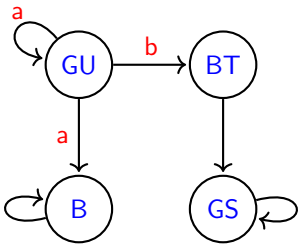


- What should a generation in GU do? a or b?

Exercises



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Should a generation in GU do a or b? The answer is: it depends. Explain what factors it might depend on.

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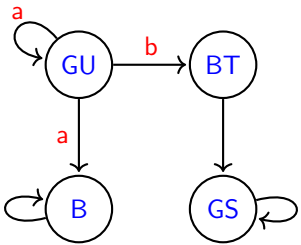

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Exercise 4.2

Assign consistent probabilities to the edges of the transition graph above.

Exercises

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Not every decision can be applied in every state. Decisions (controls) that can be applied in a given state are said to be feasible for that state. Give an example of a simple maze exit game in which certain controls are not feasible. What could be the type of a *feasible* predicate?

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Even a two-step decision plan could be infeasible. Explain why using the example from Exercise 4.3.

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Exercise 4.5

Under stochastic uncertainty, it is often not a good idea to take decisions which are optimal for the expected value. Explain why. Give an example in which this is in fact the worst that one can do!

Examples 1 and 2: common traits

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- The consequences of these decisions are uncertain
- Uncertainties can be of different kinds (remember OUU!)
- Making a good decision at a given step requires accounting for all its immediate possible consequences and for all possible consequences of future decisions

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- Making good decisions in SDPs (optimal planning) requires understanding a simple but non-trivial fact
- When the evolution of a system is uncertain, the notion of an optimal sequence (plan, path) of decisions (controls) becomes problematic
- This is because, under uncertainty, more than one evolution is possible
- One has to compute a best decision (control) for all states that may occur after that decision: a policy

Lecture 2: specifying and solving SDPs under uncertainty

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